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Original Research

No increase in adverse events with lateral extra-articular tenodesis augmentation of anterior cruciate ligament reconstruction – Results from the stability randomized trial



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ARTICLE INFO

Keywords:

Anterior cruciate ligament

ACL

Knee

Lateral extra articular tenodesis

LET

Adverse events

ACL reconstruction

ABSTRACT

Objectives: Results from the Stability Study suggest that adding a lateral extra-articular tenodesis (LET) to a hamstring tendon autograft reduces the rate of anterior cruciate ligament reconstruction (ACLR) failure in high-risk patients. The purpose of this study is to report adverse events over the 2-year follow-up period and compare groups (ACLR alone vs. ACLR + LET).

Methods: Stability is a randomized clinical trial comparing hamstring tendon ACLR with and without LET. Patients aged 14–25 years with an ACL deficient knee were included. Patients were followed and adverse events documented (type, actions taken, resolution) with visits at 3, 6, 12, and 24 months postoperatively. Adverse events were categorized as none, minor medical, minor surgical, contralateral ACL rupture, or graft rupture. Patient reported outcome measures (PROMs) collected at each visit included the Knee Injury and Osteoarthritis Outcome Score (KOOS), International Knee Documentation Committee Score (IKDC), and ACL Quality of Life Questionnaire (ACL-QOL).

Results: In total, 618 patients were randomized (mean age 18.9 years, 302 (49%) male). Forty-five patients (7%) suffered graft rupture; 34 (11%) in the ACLR group compared to 11 (4%) in the ACLR + LET group (RRR = 0.67, 95% CI 0.36 to 0.83, $p < 0.001$). There were no differences in effusion or infection rates between groups. The ACLR + LET group experienced an increased number of hardware removals (10 vs. 4). Overall, the rate of minor medical events (11%), minor surgical events (7%), and ipsilateral or contralateral ACL tears (10%) were low considering the high-risk patient profile. Increasing severity of adverse events was associated with lower PROMs at 24 months post-operative. Patients in the ACLR + LET group reported greater degree of pain at 3 months only. There were no clinically significant differences in range of motion between groups.

Conclusions: The addition of LET to hamstring tendon autograft ACLR in young patients at high risk of re-injury resulted in a statistically significant reduction in graft rupture. While the addition of LET may increase rates of hardware irritation, there was no significant increase in overall rates of minor medical adverse events, minor

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<https://doi.org/10.1016/j.jisako.2022.12.001>

Received 10 August 2022; Received in revised form 2 December 2022; Accepted 8 December 2022

Available online 14 January 2023

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surgical events, or overall re-operation rates. The concerns regarding complications associated with a LET did not materialize in this study.
Level of evidence: Level I

What are the new findings?

We found a clinically relevant reduction in anterior cruciate ligament graft rupture with the addition of a lateral extra-articular tenodesis (LET) in young patients at high risk of re-injury. The addition of a LET did not increase the rate of serious adverse events and reduced the rates of new meniscal tears. The addition of a LET did increase the incidence of hardware removal due to irritation at the LET repair site.

Introduction

Recent research has demonstrated that anterior cruciate ligament reconstruction (ACLR) does not always result in optimal outcomes. Studies have identified a high rate of persistent rotatory laxity as demonstrated by a positive pivot shift [1,2], reduced levels of return to sport [3], and high rates of failure. The latter has been demonstrated to be worse with younger age [4,5], probably because age is a surrogate for activity level [6].

Significant focus has recently been placed on the anterolateral complex (ALC), particularly in regard to anterolateral ligament (ALL) reconstruction or lateral extra-articular tenodesis (LET), as a means to provide greater rotational stability following ACLR, with the hope of reducing the failure rate [7]. The LET has been utilized in knee surgery to address ACL deficiency for many decades. While a number of studies showed improved stability rates when combined with ACLR [8], it fell out of favour following an American Orthopaedic Society for Sports Medicine (AOSSM) consensus meeting that determined that the perceived high rates of complications observed did not make the procedure worth performing [9]. The main concerns cited were lateral compartment over constraint, osteoarthritis, and irritation from disturbing the iliotibial band.

With the goal of reducing ACLR failure, we completed a multicenter randomized clinical trial (Stability Study) to evaluate single bundle, hamstring autograft ACLR with LET in young patients deemed to be at high risk of graft failure [10]. The addition of a LET to the hamstring autograft ACLR resulted in a clinically important and statistically significant reduction in the primary outcome of clinical failure (composite of persistent rotatory laxity and graft rupture) (40% vs. 25%; relative risk reduction (RRR), 0.38; 95% CI, 0.21–0.52; $p < 0.0001$) [10].

Due to the historical concern of associated morbidity with the addition of LET, the purpose of this paper is to report the adverse event profile within the Stability Study and compare the rates of adverse events between those who had ACLR with and without LET. Our hypothesis was that with modern surgical and rehabilitation techniques there would be no increase in complication rates with the addition of the LET.

Methods

Study design and participants

This study was a pragmatic, parallel groups, multicenter randomized clinical trial in which young patients with ACL deficiency were randomly allocated to either ACLR or ACLR with LET (ACLR + LET). Seven study centres in Canada and two centres in Europe actively recruited patients. The study was approved by Western University's Research Ethics Board and local Research Ethics Boards at each institution and was registered

on Clinical Trials.gov (NCT02018354). A full study protocol and primary outcome results have been previously published [11,10].

Briefly, patients were approached for participation if they were between 14 and 25 years old, had an ACL deficient knee, and were at a higher risk of re-injury, defined as the presence of two or more of the following: (1) participating in competitive pivoting sport; [12] (2) presence of a grade 2 pivot shift or greater; (3) generalized ligament laxity (Beighton score of 4 or greater) or genu recurvatum greater than 10 degrees [13]. Patients were ineligible if any of the following were present: (1) previous ACL reconstruction on either knee; (2) multiligament injury (two or more ligaments requiring surgical attention); (3) a symptomatic articular cartilage defect requiring treatment other than debridement; (4) greater than 3 degrees of asymmetric varus; and (5) unable or unwilling to be followed up for 2 years post-operative.

Upon determining eligibility, and willingness to participate, patients were randomized via phone or web-based software program (Empower Inc.) in a 1:1 ratio to either ACLR alone or ACLR with LET, stratified by surgeon, sex and meniscus repair. All patients underwent an anatomic single bundle hamstring autograft ACLR performed in a standardized fashion across sites. Patients who were randomized to receive LET had this performed using a standardized modified Lemaire technique, as previously published [14].

All patients, regardless of group allocation, received preoperative and postoperative verbal and written standardized instructions for rehabilitation, focused on early range of motion and weight bearing as tolerated, unless a meniscus repair dictated otherwise. The patient's physical therapist also received a copy of the standardized protocol. A brace was not routinely used.

Adverse events and patient-reported outcomes

Subjects were evaluated at 2 and 6 weeks, and 3, 6, 12, and 24 months following surgery. An adhesive tape was applied to the lateral side of the knee concealing presence or absence of LET scar to ensure that outcome assessors were blind to group allocation.

Any adverse event that occurred during the study period was recorded prospectively. For each adverse event, the following were reported: identification (general, anaesthesia related, wound related, neurovascular, knee specific, procedure related), actions taken (medications, surgery, etc.) and whether the event had resolved. Graft rupture was defined as a tear of the graft confirmed by either MRI or arthroscopic examination (performed in response to suspicion of a graft rupture).

Range of motion was assessed by measuring passive knee extension and active-assisted knee flexion using a standardized approach with a goniometer. Pain was recorded using the P4 questionnaire, which consists of four items that address pain intensity in the morning, afternoon, evening, and with activity over the past 2 days [15,16]. Each item is scored on a 0-to-10 numeric pain rating scale; therefore, the total P4 score can vary from 0 (no pain) to 40 (the highest possible pain level).

Patient-reported outcome measures (PROMs) included the ACL Quality of Life Questionnaire (ACL-QOL) [17], International Knee Documentation Committee Subjective Knee Form (IKDC) [18], and the Knee Injury and Osteoarthritis Outcome Score (KOOS) [15]. All three scores range from 0 (worst) to 100 (best).

Sample size calculation

The primary outcome of the Stability Study was graft failure, as defined by either a graft rupture or symptomatic instability requiring

revision ACLR surgery, or persistent rotational laxity as measured by an asymmetrical positive pivot shift (grade 2 pivot shift or higher at any post-operative visit, or grade 1 pivot shift at consecutive post-operative visits) compared to the contralateral side. Based on this primary outcome, we estimated that within this group of high-risk patients, the absolute risk (AR) of graft failure (as defined above) following ACLR ranges from 25 to 35% [19,20,1,4]. We agreed that a relative risk reduction (RRR) in graft failure rate of at least 40% would merit a change in practice (i.e. is of sufficient magnitude to warrant the additional costs related to the LET procedure). Thus, with 255 patients per group and a type 1 error rate of 5% there would be approximately 80% power to detect a relative risk reduction (RRR) in rate of re-rupture in the ACLR + LET group of 40% or greater. A combined withdrawal and lost-to-follow-up rate of approximately 15% was anticipated, requiring a total recruitment of 600 patients (300 per group).

Statistical analysis

All patients were analysed in the group to which they were randomized (i.e., intention to treat principle). For graft rupture in each group, the absolute risk, relative risk (RR), and risk difference (RD) of graft rupture with 95% confidence intervals around each estimate, were calculated using a Mantel Haenszel Test (random effect of surgeon) to determine the significance of the association between the addition of LET and graft failure rates.

We report descriptive information (number and proportion of patients) for the specific adverse events (general, related to ACLR, related to LET). To statistically compare the adverse event profile between groups (ACLR alone vs. ACLR + LET), we categorized the level of adverse event patients experienced into four groups: none (no adverse event), minor medical (event such as a superficial infection, pain, or muscle strain that either resolved spontaneously or with minimum medical management), minor surgical (adverse event such as meniscus tear or stiffness that required surgical intervention but is not a graft rupture), contralateral ACL rupture, and graft rupture. Each patient was only included once, so patients with more than one adverse event were included in the category of their most severe adverse event. We reported the proportion of patients with each adverse event type and conducted a Pearson chi-square test to make statistical comparisons between groups.

For the P4, the mean and standard error for each group at each time point was calculated with the mean between-group difference with 95% confidence interval at 1 and 2 years postoperative. An analysis of covariance was also completed where the preoperative (baseline score) was used as a covariate.

To investigate the association of adverse events to P scores, we used a generalized linear model comparing mean PROM scores reported for patients with the different levels of adverse events (none, minor medical, minor surgical, graft/ACL tear). We performed multiple imputation using chained equations (MICE) using potential predictors of missingness and outcome (age, sex, BMI, group, adverse event category, baseline PROM scores, and 12-month PROM scores) to impute 24-month outcome scores for those missing PROMs. We performed five iterations, conducted diagnostic evaluations for each iteration, then used the pooled data sets in our analysis. We used a generalized linear model adjusted for baseline questionnaire scores, to determine the effect of adverse event type on 24-month outcome scores. We present pooled adjusted mean scores and standard errors for each adverse event group and report p-values representing the significance of the difference in means between each adverse event group and the group with no adverse events. We performed a sensitivity analysis where we removed patients that had an adverse event at the 24-month timepoint to determine whether the effects of these events were due to the recency of the event rather than the severity itself. Adverse events reported after 24-month PROMs had been completed ($n = 4$) were not included in this analysis but were included in the descriptive data. All analyses were performed in SPSS (IBM version 28) and statistical significance was set at $p < 0.05$.

Results

Subject characteristics

Enrolment of subjects took place between January 2014 and March 2017. Of the 1033 subjects screened for eligibility (Fig. 1), 358 were ineligible and 48 declined participation. Thus, 618 patients were randomized; 589 completed the study, 18 patients were lost to follow-up, and 11 patients were withdrawn (5% attrition rate). There were no statistical differences between groups for any patient characteristics at baseline (see Table 1).

Adverse events

The adverse event status was documented for all 618 randomized patients at surgery and follow-up visits until they completed the study at 24 months post-operative, were lost to follow-up, or withdrawn. Table 2 shows a detailed distribution of adverse events reported in the study up to 2 years post-operative. In total, 45 patients experienced graft rupture, 34 of 312 (11%) in the ACLR group compared with 11 of 306 (4%) in the ACL + LET group (RRR, 0.67; 95% CI, 0.36–0.83; $p < 0.001$). The RD was 8% (95% CI, 3%–12%).

There was no difference in the rates of persistent effusion, superficial or deep infection. The addition of the LET saw an increased number of procedure-related complications, with fewer patients complaining of hardware irritation in the ACLR than ACLR + LET group (4 vs. 14). Six LET graft difficulties were reported at the time of surgery. These included three cases of LET graft rupture, and three cases of LET staple hardware failure resolved with hardware exchange. Ten of the ACLR + LET patients required removal of the LET staple.

There was no significant difference in the proportion of patients experiencing minor medical or minor surgical adverse events, or contralateral ACL tears between groups (Table 3). A significantly larger proportion of patients in the ACLR alone group experienced a graft rupture (11%) compared to the ACLR + LET group (4%, $p < 0.01$). The overall rate of reoperation was not significantly different between the ACLR alone (18%) and ACLR + LET (15%) groups ($p = 0.37$).

Adverse event data for the entire sample, including frequencies of each type, are shown in Table 4. Overall, 70% of minor medical complications (48 of 69) and 63% of minor surgical complications (29 of 46) reported in the 2-year study period occurred within the first 12-months post-operative, compared to 37% of contralateral ACL tears or 49% (7 of 19) of graft ruptures (22 of 45). About half of the overall reoperations occurred in each of the first and second year post-operative.

Pain

For both groups, pain was minimal by 3-months post-operative (approximately 8/40 on the P4). Overall pain was lower in the ACLR compared to the ACL + LET group (adjusted mean difference = -1.7 (95% CI -2.9 to -0.6), $p = 0.004$) (Table 5). This difference was not observed beyond 3-months post-operative.

Range of motion

No clinically significant differences in range of motion were observed between the two treatment groups at 6, 12, or 24 months post-operative. At 3-months post-operative, the ACLR + LET group demonstrated a 2.1° (95% CI 0.6 to 3.6) reduction in flexion and a 0.7° (1.3–0.1) reduction in extension compared to ACLR alone (Table 6).

Patient-reported outcome measures

Complete questionnaire data were available for 534 patients (86.4%) at 2-years post-operative, and we imputed scores for the remainder. Sixteen patients missing baseline data were excluded from the analysis.

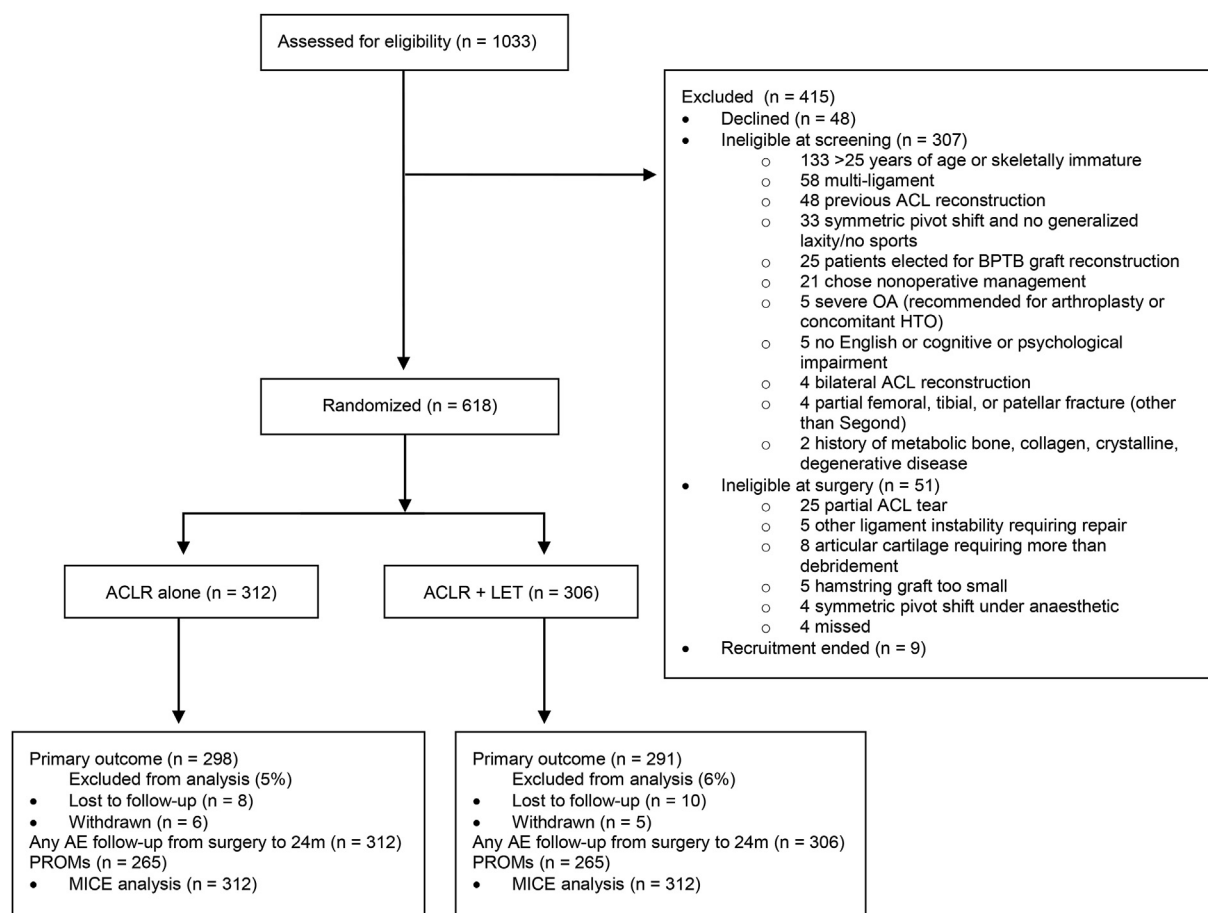


Fig. 1. Consolidated Standards of Reporting Trials (CONSORT) flow diagram for the Stability Study. ACL, anterior cruciate ligament; ACLR, anterior cruciate ligament reconstruction; AE, adverse event; BPTB, bone-patellar tendon-bone; HTO, high tibial osteotomy; LET, lateral extra-articular tenodesis; MICE, multivariate imputation by chained equations; OA, osteoarthritis; PROM, patient-reported outcome measure.

When all complications recorded up to 24-months post-operative were included, 24-month PROM scores for patients with any type of complication were significantly lower ($p < 0.05$) than those with no complications across all outcomes except for the KOOS ADL subscale (Table 7). For each of the significant outcomes, there was a trend of lower pooled mean scores as complication severity increased. The sensitivity analysis, which removed patients with complications that occurred between 12 and 24 months ($n = 73$), showed that graft tears and contralateral ACL tears within the first 12-months post-operative led to significantly lower PROM scores at 2-years post-operative ($p < 0.05$), while the effect of smaller minor surgical complications did not always have the same impact (Table 8). Minor surgical complications within the first 12-months postoperative did lead to statistically significant differences in outcome score on the ACL-QOL and KOOS Pain subscale ($p < 0.05$). In this analysis, the pooled mean scores for patients with minor complications were not significantly different than those with no complications for any outcome, and surgical complications only showed significant differences from the no complications group in ACL-QOL and KOOS Pain scores.

Discussion

The most important finding from this study was that the addition of LET to a hamstring tendon autograft ACLR in patients under the age of 25 years significantly reduces graft rupture rates compared to ACLR alone (4% vs. 11%), with no significant increase in any associated serious adverse events. When adverse events were categorized, there was a significantly lower proportion of patients with graft or contralateral ACL

tears in the LET group. Additionally, there was no significant difference in the proportion of patients with minor medical adverse events (which resolved spontaneously), other minor surgical adverse events (i.e., meniscal tears, stiffness, hardware irritation and removal), or overall reoperation rates between those who had the ACLR alone and those with ACLR + LET.

We previously reported that the difference in graft failure between groups did not result in between group differences in the PROMs collected at 6, 12, and 24 months [10]. However, the analysis from this paper clearly shows that a more severe complication such as graft rupture or contralateral ACL injury does result in significantly reduced PROMs. In contrast, the effect of minor medical and minor surgical complications was washed out when patients with 24-month complications were excluded. This indicates that the negative influence of these complications on PROMs in our current analysis was likely due to the recency of the event rather than the severity of the complication. However, patients who experience a graft rupture or contralateral ACL tear appear to have a significantly lower PROMs, regardless of the time at which the tear occurs. For this analysis, we grouped graft rupture and contralateral ACL tears into one composite outcome (graft/ACL tear) to increase the statistical power to investigate the influence of having a secondary ACL injury event on patient-reported outcome scores.

Furthermore, we did not see an increased rate of septic arthritis to be observed with the addition of LET, as has been suggested in a previous study [16]. We did however observe a significant increase in pain up until 3-months post-operative with the addition of the LET, likely as a result of the extra surgery performed. Importantly, these differences resolved by 6-months post-operative.

Table 1

Patient demographics. ACLR, anterior cruciate ligament reconstruction; LET, lateral extra-articular tenodesis; BMI, body mass index; GLL, generalized ligamentous laxity; RTS, return to sport; ICRS, International Cartilage Repair Society.

Characteristic	ACLR alone	ACLR + LET	p value
Sex, n males (%)	151 (48)	151 (49)	0.44
Age, years (mean ± SD)	18.8 ± 3.2	19.1 ± 3.3	0.33
Height, inches (mean ± SD)	68.1 ± 3.7	67.8 ± 3.6	0.29
Weight, kg (mean ± SD)	71.8 ± 14.8	71.3 ± 14.3	0.75
BMI, kg/m ² (mean ± SD)	23.8 ± 3.7	24.0 ± 3.8	0.68
Beighton score, 0–9 (mean ± SD)	3.1 ± 2.7	3.0 ± 2.8	0.58
Eligibility Criteria			
+GLL, +pivot*, +sport	125 (40)	124 (41)	0.37
+GLL, +pivot*, –sport	11 (4)	9 (3)	
+GLL, –pivot*, +sport	28 (9)	17 (6)	
–GLL, +pivot*, +sport	144 (47)	153 (50)	
Time from injury to surgery, months (mean ± SD)	8.1 ± 18.9	9.3 ± 16.7	0.41
Operative limb, n dominant (%)	161 (52)	156 (52)	0.98
Mechanism of injury, n non-contact (%)	176 (74)	166 (72)	0.32
Sport played at time of injury, n (%)			0.06
Soccer	100 (32)	122 (39)	
Basketball	54 (18)	36 (12)	
Football or Rugby	54 (18)	56 (19)	
Downhill skiing	16 (5)	13 (4)	
Volleyball	19 (6)	12 (4)	
Other	66 (21)	66 (22)	
Sport played at time of RTS, n (%)			
Soccer	72 (23)	81 (26)	
Basketball	37 (12)	27 (9)	
Football or Rugby	27 (9)	30 (10)	0.49
Downhill skiing	13 (4)	10 (3)	
Volleyball	16 (5)	15 (5)	
Other	99 (32)	87 (28)	
Did not return to sport	47 (15)	56 (18)	
Smoking status, n (%)			
Current smoker	17 (5)	16 (5)	
Ex-smoker	11 (4)	6 (2)	
Non-smoker	279 (91)	279 (93)	
Graft source, n (%)			0.57
Semitendinosus and gracilis tendons	301 (96)	297 (96)	
3 strand	2 (<1)	2 (<1)	
4 strand	140 (45)	135 (44)	
5 strand	119 (38)	119 (39)	
6 strand	38 (12)	39 (13)	
Semitendinosus tendon	11 (4)	11 (4)	
3 strand	5 (2)	3 (1)	
4 strand	6 (2)	8 (3)	
Graft diameter, mm (median, min, max)	8 (6, 10)	8 (6, 10)	0.32
Meniscectomy, n (%)			
Medial	21 (7)	18(6)	
Lateral	67 (22)	54 (18)	
Both	11 (4)	12 (4)	
Meniscal repair, n (%)			
Medial	75 (24)	91 (20)	
Lateral	36 (12)	24 (8)	
Both	19 (6)	18 (6)	
Change in rehab due to meniscus repair, n (%)	51 (16)	50 (16)	
Chondral defect, ICRS >3 any compartment, n (%)	15 (5)	14 (4)	0.52
Time (months) to RTS, median (IQR)	11 (8–14)	11 (8–17)	0.01
Exposure to sport (months), median (IQR)	13 (4–16)	13 (5–15)	0.40

Note: *under anaesthetic.

ACLR alone: n = 312.

ACLR + LET: n = 306.

Similar to previous studies, this large, randomized trial found a higher rate of hardware related discomfort in the ACLR + LET group, ultimately requiring removal of the staple. This secondary surgery should be placed in the context of the reduced rate of graft rupture, as well as total re-operation rates. It is possible that the issue of LET staple removal could be mitigated with use of an alternative fixation device; however, this might negatively influence the cost effectiveness of the procedure and may impact efficacy. Despite 10 patients in the LET group having a re-operation for LET staple

Table 2

Adverse events by group. ACLR, anterior cruciate ligament reconstruction; FCL, fibular collateral ligament; LET, lateral extra-articular tenodesis; MUA, manipulation under anaesthesia; PT, physiotherapy.

General	ACLR	ACLR + LET	Overall
Persistent effusion	6 (2%)	10 (3%)	16 (3%)
Aspiration	3 (1%)	1 (<1%)	4 (<1%)
Reoperation (arthrolysis, resect adhesions, cyclops)	2 (<1%)	1 (<1%)	3 (<1%)
Superficial infection	2 (<1%)	4 (1%)	6 (1%)
Deep infection	2 (<1%)	1 (<1%)	3 (<1%)
Periostitis	0	1 (<1%)	1 (<1%)
Related to ACLR	ACLR	ACLR + LET	Overall
Graft rupture	34 (11%)	11 (4%)	45 (8%)
Contralateral ACL rupture	12 (4%)	7 (2%)	19 (3%)
Excessive pain	6 (2%)	6 (2%)	12 (2%)
Patellofemoral (physio)	0	1 (<1%)	1 (<1%)
Intra-articular injection	2 (<1%)	1 (<1%)	3 (<1%)
Reoperation (scar resection, cyclops)	1 (<1%)	1 (<1%)	2 (<1%)
Excessive stiffness (all MUA, scar resection, cyclops, arthrolysis)	2 (<1%)	5 (2%)	7 (1%)
ACL hardware removal	4 (1%)	1 (<1%)	5 (<1%)
Hamstring strain/tear	6 (2%)	2 (<1%)	8 (1%)
Retear meniscal tear (unrelated to graft rupture)	7 (2%)	13 (4%)	20 (3%)
Retear meniscal tear (associated with a graft rupture)	0	1 (<1%)	1 (<1%)
New meniscal tear (unrelated to graft rupture)	6 (2%)	3 (<1%)	9 (2%)
New meniscal tear (associated with a graft rupture)	2 (<1%)	1 (<1%)	3 (<1%)
Complex regional pain syndrome	0	1 (<1%)	1 (<1%)
MCL (1 grade 1 sent for PT, 1 rupture reoperation to re-insert)	0	2 (<1%)	2 (<1%)
Related to LET	ACLR + LET	Overall	
Intraoperative			
LET graft difficulties at surgery	6 (2%)	6 (1%)	
Damage to FCL attachment (repaired)	1 (<1%)	1 (<1%)	
Postoperative			
Haematoma over LET site	3 (<1%)	3 (<1%)	
IT band snapping	2 (<1%)	2 (<1%)	
LET hardware removal	10 (3%)	10 (2%)	
Over-constrained lateral compartment	1 (<1%)	1 (<1%)	

An overconstrained lateral compartment was defined subjectively based on reduced internal rotation as assessed by a surgeon and reduced motion as assessed by the patient.

ACLR alone: n = 312.

ACLR + LET: n = 306.

Table 3

Adverse event category by group. ACLR, anterior cruciate ligament reconstruction; LET, lateral extra-articular tenodesis.

Adverse event category	ACLR	ACLR + LET	p-value
Minor medical adverse events	29 (9%)	40 (13%)	0.14
Minor surgical events (excluding ACL tears)	18 (6%)	28 (9%)	0.11
Contralateral ACL rupture	12 (4%)	7 (2%)	0.26
Graft rupture	34 (11%)	11 (4%)	<0.01
Overall reoperation rate	56 (18%)	46 (15%)	0.32

Bolded values indicate significant difference in proportion of event between groups.

ACLR alone: n = 312.

ACLR + LET: n = 306.

removal, the overall rate of re-operation between groups was similar between groups and slightly higher for the ACLR alone group (18%) versus the ACLR + LET group (15%), likely due to the higher rates of graft rupture in the former.

Table 4

Adverse events recorded within the first 12-months post-operative and the total number of adverse events recorded within the entire 24-month period. ACLR, anterior cruciate ligament reconstruction; LET, lateral extra-articular tenodesis; ACL, anterior cruciate ligament.

	12-months post-operative	24-months post-operative
No Adverse Events	512 (83%)	439 (71%)
Minor Medical Adverse Events	48 (8%)	69 (11%)
ACLR	18 (6%)	29 (9%)
ACLR + LET	30 (10%)	40 (13%)
Minor Surgical Events (excluding ACL tears)	29 (5%)	46 (7%)
ACLR	11 (4%)	18 (6%)
ACLR + LET	18 (6%)	28 (9%)
Contralateral ACL Tear	7 (1%)	19 (3%)
ACLR	6 (2%)	12 (4%)
ACLR + LET	1 (<1%)	7 (2%)
Graft Rupture	22 (4%)	45 (7%)
ACLR	18 (6%)	34 (11%)
ACLR + LET	4 (1%)	11 (4%)
Overall Reoperation Rate	54 (9%)	102 (16%)
ACLR	32 (10%)	56 (18%)
ACLR + LET	22 (7%)	46 (15%)

In a study by Sonnery-Cottet et al., they observed that the addition of an LET seemed to be protective of meniscal repair [21]. This may be attributable to the improved kinematics of the knee, that have been shown in cadaveric studies when lateral procedures are added to an ACL reconstruction [22]. However, the rate of new mensical tears and re-tears was not different between groups in our study. Furthermore, concerns of lateral over constraint and lateral compartment osteoarthritis are often suggested to be associated with lateral procedures. The evidence to

Table 5

Patient reported pain for each group over time. ACLR, anterior cruciate ligament reconstruction; LET, lateral extra-articular tenodesis; MD, mean between groups difference; m, months.

P4 Score (0–10)	ACLR alone	ACLR + LET	Adjusted MD (95% CI)	p-value
3 m	6.8 ± 0.4	8.6 ± 0.4	−1.7 (−2.9 to −0.6)	0.004
6 m	4.7 ± 0.4	5.3 ± 0.4	−0.6 (−1.6 to 0.5)	0.27
12 m	3.4 ± 0.3	3.4 ± 0.3	0.0 (−0.9 to 0.9)	1.0
24 m	2.9 ± 0.3	3.0 ± 0.4	−0.1 (−1.1 to 0.9)	0.84

Bolded values indicate significant differences between groups.

ACLR alone: *n* = 312.

ACLR + LET: *n* = 306.

Table 6

Range of motion on the operative side, side-to-side difference, and adjusted mean difference between groups for side-to-side difference. ACLR, anterior cruciate ligament reconstruction; LET, lateral extra-articular tenodesis.

ROM	ACL alone		ACL + LET		Adjusted mean difference (95% CI)	p-value
	Operative	Side-to-side	Operative	Side-to-side		
Passive extension						
Baseline	−1.9 ± 0.4	1.5 ± 0.2	−2.8 ± 0.3	1.3 ± 0.2	0.2 (−0.4 to 0.8)	0.43
3 month	−1.4 ± 0.3	2.6 ± 0.2	−0.8 ± 0.3	2.7 ± 0.2	−0.7 (−1.3 to −0.1)	0.03
6 month	−2.9 ± 0.3	1.3 ± 0.2	−2.7 ± 0.3	1.6 ± 0.2	−0.3 (−0.8 to 0.2)	0.28
12 month	−3.0 ± 0.3	0.8 ± 0.2	−3.2 ± 0.3	0.9 ± 0.2	−0.1 (−0.5 to 0.3)	0.63
24 month	−3.1 ± 0.3	0.6 ± 0.1	−3.2 ± 0.3	0.8 ± 0.1	−0.2 (−0.6 to 0.2)	0.27
Active assisted flexion						
Baseline	137.5 ± 0.6	−4.1 ± 0.5	138.9 ± 0.6	−3.1 ± 0.5	−0.9 (−2.2 to 0.4)	0.16
3 month	135.0 ± 0.7	−5.3 ± 0.5	134.2 ± 0.7	−7.4 ± 0.6	2.1 (0.6 to 3.6)	<0.01
6 month	138.9 ± 0.5	−3.3 ± 0.4	137.9 ± 0.6	−4.3 ± 0.4	1.0 (−0.0 to 2.1)	0.05
12 month	139.9 ± 0.5	−2.5 ± 0.3	139.7 ± 0.5	−2.2 ± 0.3	−0.3 (−1.2 to 0.7)	0.58
24 month	140.6 ± 0.6	−2.5 ± 0.3	140.4 ± 0.6	−2.2 ± 0.3	−0.3 (−1.2 to 0.6)	0.53

Bolded values indicate significant differences between groups.

ACLR alone: *n* = 312.

ACLR + LET: *n* = 306.

Note: Negative values for the operative side indicate hyperextension. Positive values for side-to-side differences in extension indicate a loss of extension on the operative side. Negative values for side-to-side differences in flexion indicate a loss of flexion on the operative side.

support the genesis of osteoarthritis being associated with meniscal loss far outweighs that of the anecdotal concerns of OA development being associated with LET. Two European studies with greater than 20 years follow up have not demonstrated an increased rate of OA development with the addition of LET and a [9,5]meta-analysis also found no correlation between LET and OA [23,24,25].

We suggest that the benefits of greater knee stability, reduced graft rupture and protection of meniscal integrity that is afforded to ACLR when an LET is added, far outweighs any transient increase in procedure related knee pain or hardware removal. However, we do recognize the need for longer-term follow-up of our patients to ensure that no differences in lateral compartment degenerative changes are observed.

Importantly, two systematic reviews in 2021 have indicated the benefits of lateral procedures being added to ACLR [26,27]. Kunze and colleagues' review of 46 studies reported that while the evidence was heterogenous, ALL reconstruction improved pivot shift outcomes (post-operative knee stability) and resulted in comparable clinical and functional outcomes the ACLR alone [27]. Beckers and colleagues reviewed 11 randomized controlled trials and prospective cohort studies which suggested that additional lateral augmentation reduced rates of graft rupture (3% vs. 12%) and rotational laxity (6% vs. 14%) as compared to isolated ACLR [26]. Both groups had similar IKDC and Tegner scores, while better Lysholm scores were reported with lateral augmentation [26]. Several studies investigating lateral extra articular procedures have been published in the past year since these reviews. Two retrospective studies reported significantly lower rates of revision in young athletes who had ACLR with two different versions of LET (Arnold Coker modification to MacIntosh procedure and Lemaire procedure) versus isolated ACLR [28,29]. A cohort study from the SANTI study group of over 300 athletes reported athletes under 21 years old who had isolated ACLR versus those who had a lateral extra-articular procedure [30]. Lastly, a 2022 ISAKOS consensus statement reached an unanimous agreement that adding LET to ACLR in particular cases (active young (<25) athletes, hyperlaxity, increased rotatory laxity, and revisions) could be associated with an increased return to sport rate in soccer [31].

This study has limitations. First, patients were not blinded to their treatment allocation. This was not possible due to the location of skin incisions, however, this was mitigated somewhat by the blinding of outcome assessors. Second, this study only investigated patients who underwent ACLR with a hamstring tendon autograft. The use of a bone patella tendon bone autograft may confer different results [1,32]. Further, we did make attempts to ensure that graft size was at least 8 mm in diameter which is in line with recent data suggesting that grafts less than 8 mm are associated

Table 7

24-month PROM scores for patients with various levels of adverse events occurring up to 24-months post-operative ($n = 602$), represented as mean $\pm$ SE (p-value). ACL-QOL, ACL Quality of Life Questionnaire; IKDC, International Knee Documentation Committee Score; KOOS, Knee Injury and Osteoarthritis Outcome Score; ADLs, Activities of Daily Living.

	None	Minor medical	Minor surgical	Graft/ACL tear
ACL-QOL	81.1 $\pm$ 1.0 (ref)	71.9 $\pm$ 2.4 (<0.01)	66.7 $\pm$ 3.4 (<0.01)	55.4 $\pm$ 2.6 (<0.01)
IKDC	89.2 $\pm$ 0.7 (ref)	83.5 $\pm$ 1.6 (<0.01)	81.0 $\pm$ 2.1 (<0.01)	71.1 $\pm$ 1.9 (<0.01)
KOOS pain	93.0 $\pm$ 0.5 (ref)	89.8 $\pm$ 1.2 (0.02)	87.0 $\pm$ 1.6 (<0.01)	84.7 $\pm$ 1.7 (<0.01)
KOOS symptoms	86.2 $\pm$ 0.7 (ref)	82.2 $\pm$ 1.6 (0.02)	79.4 $\pm$ 2.3 (<0.01)	75.0 $\pm$ 2.3 (<0.01)
KOOS ADLs	97.4 $\pm$ 0.3 (ref)	95.2 $\pm$ 0.8 (0.01)	95.7 $\pm$ 1.1 (0.13)	92.1 $\pm$ 1.0 (<0.01)
KOOS sport	87.3 $\pm$ 0.9 (ref)	81.5 $\pm$ 2.1 (0.01)	77.4 $\pm$ 2.8 (<0.01)	67.2 $\pm$ 2.9 (<0.01)
KOOS QOL	78.1 $\pm$ 1.0 (ref)	68.9 $\pm$ 2.5 (<0.01)	66.1 $\pm$ 3.4 (<0.01)	53.5 $\pm$ 3.3 (<0.01)

p-values represent statistical significance in pooled mean scores between each complication group and the group with no complications (i.e., reference level).

SE = standard error.

Ref = reference level, no p-value available.

Table 8

24-month PROM scores for patients with various levels of adverse events complications occurring within the first 12 months post-operative ($n = 545$), represented as mean $\pm$ SE (p-value). ACL-QOL, ACL Quality of Life Questionnaire; IKDC, International Knee Documentation Committee Score; KOOS, Knee Injury and Osteoarthritis Outcome Score.

	None	Minor medical	Minor surgical	Graft/ACL tear
ACL-QOL	81.3 $\pm$ 0.9 (ref)	76.9 $\pm$ 2.7 (0.13)	70.8 $\pm$ 4.3 (0.02)	54.0 $\pm$ 3.6 (<0.01)
IKDC	89.4 $\pm$ 0.6 (ref)	86.8 $\pm$ 1.6 (0.12)	85.4 $\pm$ 2.2 (0.08)	75.1 $\pm$ 2.4 (<0.01)
KOOS pain	93.1 $\pm$ 0.4 (ref)	92.1 $\pm$ 1.3 (0.45)	89.3 $\pm$ 1.8 (0.04)	87.5 $\pm$ 2.2 (<0.01)
KOOS symptoms	86.4 $\pm$ 0.6 (ref)	85.5 $\pm$ 1.8 (0.64)	82.4 $\pm$ 2.8 (0.18)	76.7 $\pm$ 3.1 (<0.01)
KOOS ADLs	97.4 $\pm$ 0.3 (ref)	96.5 $\pm$ 0.9 (0.29)	98.0 $\pm$ 1.2 (0.65)	94.0 $\pm$ 1.2 (<0.01)
KOOS sport	87.6 $\pm$ 0.8 (ref)	84.9 $\pm$ 2.3 (0.26)	81.0 $\pm$ 3.2 (0.05)	72.4 $\pm$ 3.9 (<0.01)
KOOS QOL	78.3 $\pm$ 0.9 (ref)	72.8 $\pm$ 2.7 (0.06)	71.6 $\pm$ 4.2 (0.12)	53.0 $\pm$ 4.8 (<0.01)

p-values represent statistical significance in pooled mean scores between each complication group and the group with no complications (i.e., reference level).

SE = standard error.

Ref = reference level, no p-value available.

with increased risks of graft failure [20]. Third, because of the age limits placed on the eligibility criteria, we are uncertain whether the study findings apply to patients under the age of 14 or those over 25 years of age. However, the 14–25 age demographic is extremely important due to their level of competitive sport participation and the high incidence of graft rupture reported in this population [6,4]. Fourth, because this was a large pragmatic trial, we were unable to ensure or measure compliance to physical therapy recommendations nor standardization of rehabilitation across all study sites, although patients were provided with standardized verbal and written instructions as per the agreed upon rehabilitation protocol. Lastly, we were unable to control for the time from surgery to return to sport. We reported six complications directly related to difficulties with the LET graft during surgery. However, this may have unfairly highlighted issues related to harvesting tissue for a LET. There were likely issues related to harvesting the ACL graft which were not reported since it is not a new procedure, and such events were unrelated to the study.

Conclusion

The addition of LET to hamstring tendon autograft ACLR in young patients at high risk of graft re-injury results in a statistically significant, clinically relevant reduction in graft rupture. While the addition of LET to an ACLR may result in increased rates of hardware irritation, there was no significant increase in overall rates of minor medical adverse events, minor surgical events, or overall re-operation rates. The concerns in the literature regarding complications associated with a LET were not observed in this study.

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Author contribution

HM completed data analysis for this manuscript. All other authors on the mast head of the manuscript were members of the steering committee of the Stability Study. AG and DB were co-principal investigators (PIs), designing the study and leading the data collection, analysis, and manuscript preparation. All other authors were the respective site PIs at individual sites. All

other authors had a role in designing the study, recruiting patients, performing surgery and leading follow up. All authors contributed to the manuscript and signed off on final draft. Members of the 'Stability Study Group' were all involved in patient recruitment and follow-up.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Alan Getgood reports financial support was provided by International Society for Arthroscopic Knee Surgery and Orthopaedic Sports Medicine (ISAKOS). Alan Getgood reports a relationship with Canadian Institutes of Health Research that includes: funding grants. Alan Getgood reports a relationship with National Institutes of Health that includes: funding grants. Alan Getgood reports a relationship with Academic Medical Organization of Southwestern Ontario that includes: funding grants. Alan Getgood reports a relationship with Canadian Foundation for Innovation that includes: funding grants. Alan Getgood reports a relationship with Ontario Research Foundation that includes: funding grants. Alan Getgood reports a relationship with Smith and Nephew that includes: consulting or advisory, royalties and funding grants. Alan Getgood reports a relationship with Graymont Inc that includes: royalties. Alan Getgood reports a relationship with Ossur that includes: consulting or advisory. Alan Getgood reports a relationship with Olympus that includes: consulting or advisory. Robert Litchfield reports a relationship with Smith and Nephew that includes: consulting or advisory. Robert McCormack reports a relationship with Canadian Institutes of Health Research that includes: funding grants. Robert McCormack reports a relationship with National Institutes of Health that includes: funding grants. Robert McCormack reports a relationship with Orthopaedic Trauma Association that includes: funding grants. Robert McCormack reports a relationship with Physicians' Services Inc Foundation that includes: funding grants. Robert McCormack reports a relationship with Bioventus that includes: speaking and lecture fees. Robert McCormack reports a relationship with Pendopharm that includes: speaking and lecture fees. Robert McCormack reports a relationship with Smith and Nephew that includes: speaking and lecture fees. Robert McCormack reports a relationship with Sanofi that includes: speaking and lecture fees. Tim Spalding reports a relationship with Conmed that includes: consulting or advisory and speaking and lecture fees. Tim Spalding reports a relationship with Smith and Nephew that includes: speaking and lecture fees. Tim Spalding reports a relationship with Joint Operations that includes: speaking and lecture fees. Peter Verdonk reports a relationship with Conmed that includes: consulting or advisory and speaking and lecture fees. Peter Verdonk reports a relationship with Smith and Nephew that includes: speaking and lecture fees. Laurie Hiemstra reports a relationship with Conmed that includes: consulting or advisory and speaking and lecture fees. Hana Marmura reports a relationship with Canadian Institutes of Health Research that includes: funding grants. Hana Marmura reports a relationship with Western University Bone and Joint Institute that includes: funding grants.

Acknowledgements

The Stability Study was funded by a research award from the International Society of Arthroscopy, Knee Surgery and Orthopaedic Sports Medicine (ISAKOS). This paper was awarded the Jan Gillquist Scientific Research Award at the 2019 International Society for Arthroscopic Knee Surgery and Orthopaedic Sports Medicine (ISAKOS) Congress.

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Update

Journal of ISAKOS

Volume 8, Issue 6, December 2023, Page 513

DOI: <https://doi.org/10.1016/j.jisako.2023.09.006>



Corrigendum

Corrigendum to ‘No increase in adverse events with lateral extra-articular tenodesis augmentation of anterior cruciate ligament reconstruction – Results from the stability randomized trial’ [Journal of ISAKOS 8 (2023) 246–254]



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The authors regret there were some errors in Table 4. The article has been updated and the correct version of the table can also be found below.

Table 4

Adverse events recorded within the first 12-months post-operative and the total number of adverse events recorded within the entire 24-month period.

	12-months post-operative	24-months post-operative
No Adverse Events	512 (83%)	439 (71%)
Minor Medical Adverse Events	48 (8%)	69 (11%)
ACLR	18 (6%)	29 (9%)
ACLR + LET	30 (10%)	40 (13%)
Minor Surgical Events (excluding ACL tears)	29 (5%)	46 (7%)
ACLR	11 (4%)	18 (6%)
ACLR + LET	18 (6%)	28 (9%)
Contralateral ACL Tear	7 (1%)	19 (3%)
ACLR	6 (2%)	12 (4%)
ACLR + LET	1 (<1%)	7 (2%)
Graft Rupture	22 (4%)	45 (7%)
ACLR	18 (6%)	34 (11%)
ACLR + LET	4 (1%)	11 (4%)
Overall Reoperation Rate	54 (9%)	102 (16%)
ACLR	32 (10%)	56 (18%)
ACLR + LET	22 (7%)	46 (15%)

The authors would like to apologise for any inconvenience caused.

DOI of original article: <https://doi.org/10.1016/j.jisako.2022.12.001>.

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<https://doi.org/10.1016/j.jisako.2023.09.006>

Available online 14 October 2023

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